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Environmental and economic analysis of harnessing solar energy in building: the case photovoltaic car park in Jijel

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Abstract

Building-integrated photovoltaics (BIPV) represent a key solution for energy transition and reducing dependency on fossil fuels. This study investigates the environmental and economic impacts associated with the implementation of a grid-connected rooftop BIPV system, with a focus on assessing the sustainability of a photovoltaic installation integrated into an automated parking in Jijel, Algeria. The proposed PV system has a total capacity of 502 kWp, determined through simulation conducted using the PVsyst software. The annual electricity production of the system was found to be sufficient to fully meet the energy demand of the parking, which is estimated at 936,519.92 kWh/year, using only 53% of the building's available rooftop area. Furthermore, the embodied energy for the entire system over its expected lifespan was calculated as 3,184,496 kWh. This resulted in an Energy Payback Time of 3.4 years and a Greenhouse Gas Payback time of 1.9 years, highlighting the system's environmental efficiency. From an economic perspective, The study also explores the potential of utilizing an additional 47% (with capacity 390 kWp) of the remaining rooftop area to generate electricity for sel to Sonelgaz, it has an estimated annual revenue of 14,658,50 \$. Additionally, the cost of electricity consumed by the parking when supplied by Sonelgaz to approximately 28,114.26 \$ per year. Therefore, the integration of the PV system could result in total annual savings of approximately 37,585.90 \$. It is worth noting that the payback period is approximately 8.2 years. These results strongly support the adoption of clean energy technologies, particularly in regions with low energy tariffs.

Keywords Photovoltaic, Parking, Environmental, Economic, Embodied energy, Payback time, PVsyst

1 Introduction

In Algeria, electricity is primarily generated via non-renewable fossil-based energy sources, particularly natural gas. This reliance poses a dual risk: it threatens the future of the national economy due to the gradual depletion of this strategic resource, and it also results in significant negative environmental impacts [1]. The Renewable Energy



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Framework Directive aims for renewables to account for 27% of energy use by 2030 [2] Fig. 1. A total of 41% of energy consumption was attributed to the residential and tertiary building sectors, making it the largest consumer [3–6] of total primary energy (Aprue) contributing to roughly 19% of greenhouse gas (GHG) emissions, emphasizing the critical need to embrace sustainable energy options for electricity, heating, cooling, and hot water. Today, solar energy (photovoltaics and solar thermal systems) have become a highly significant topic due to their minimal environmental impact and increasing widespread use in certain regions (especially for isolated zones). Despite their advantages, integration remains a challenge. Solar components are often mounted externally on roofs, which can conflict with the building's design and aesthetics, leading some architects to oppose their use. To address this issue, solar systems must be integrated into building structures in a manner that is both visually appealing and economically efficient.

Growing concerns over climate change and global warming have driven a 50% increase in global renewable energy capacity in 2023. By the end of the year, photovoltaic (PV) systems reached approximately 1.6 TW of installed capacity, accounting for nearly 47.5% of total global renewable energy. This rapid growth highlights a clear global shift toward renewable and sustainable energy technologies [7].

2 Politico-economic context

The Energy Performance of Buildings Directive (EPBD) mandates the promotion of renewable energy systems (RES), particularly solar technologies, to reduce reliance on traditional fossil fuels in buildings. The notable rise in the adoption of renewable energy sources in buildings, especially photovoltaic (BIPV) and solar thermal systems (STS) is crucial for meeting this goal and is expected to expand significantly in the years ahead. This progress is backed by the updated Energy Performance of Buildings Directive (EPBD), which requires that by the next few years, buildings within the EU attain zero or very low energy consumption levels. PV and STS systems are expected to play a key role in meeting these requirements, providing electricity, heating, cooling, and domestic hot water.

The Earth's surface is exposed to an average solar irradiance of 1366 W/m^2 , known as the solar constant. As this radiation traverses the atmosphere, it undergoes attenuation due to

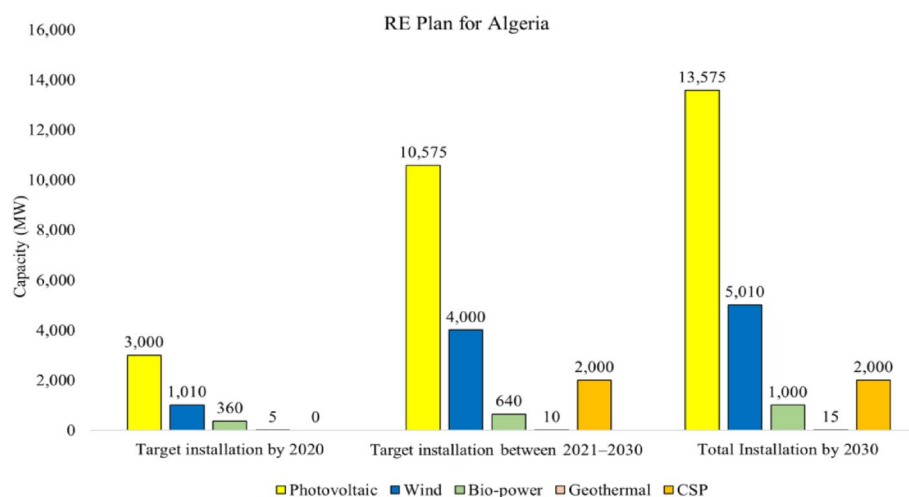
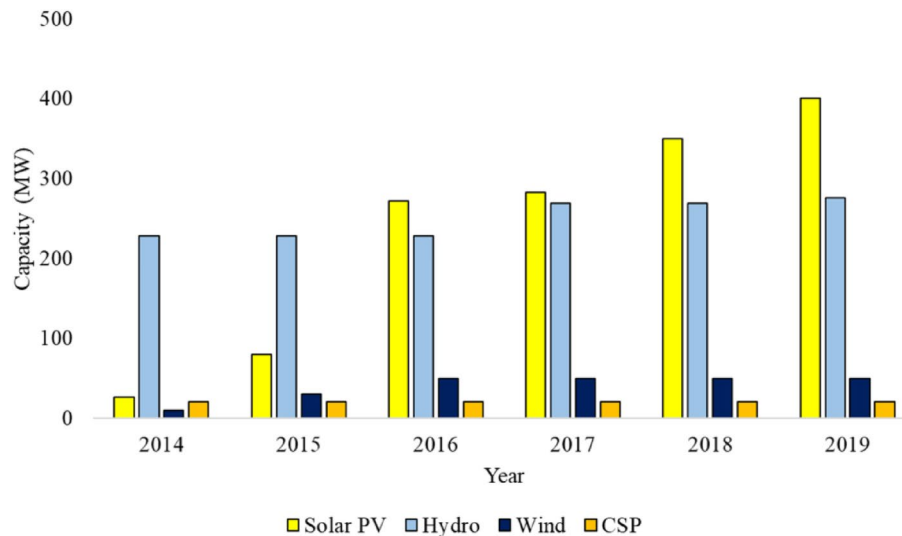


Fig. 1 Algeria RE current and future generation plan [2]

Table 1 Geographic zones and solar resource characteristics in Algeria [6]

Geographic zones	Coastal area	Inner area	Desert area
Land area distribution (%)	4	10	86
Average of the annual sunrise (hour/year)	2650	3000	3500
Mean solar radiation (kWh/m ² /year)	1700	1900	2650

**Fig. 2** Algeria's Renewable energy capacity from 2014 to 2019 [2].

absorption and scattering, yielding a maximum irradiance of approximately 1000 W/m^2 at sea level under clear-sky conditions [4]. This highlights the sun's immense potential as a sustainable and abundant energy resource, providing notable environmental advantages over conventional energy systems while also offering specific technical benefits relative to other renewable energy technologies, contingent upon geographic and climatic factors.

Algeria as situated within the Sunbelt region, it exhibits exceptional potential for leveraging solar energy due to its favorable solar resource availability. In 2011, the Algerian Ministry of Energy and Mining (MEM) enacted Law No. 11–11 to promote and financially support renewable energy (RE) projects, thereby establishing foundational measures for subsequent RE initiatives and sustainable development efforts. This policy framework acts as a pivotal incentive mechanism aimed at mitigating reliance on conventional energy systems, which are associated with elevated CO_2 emissions and susceptibility to fluctuations in global oil and gas market prices [5]. Our country receives an estimated $169,440 \text{ kW/m}^2$ per year of direct solar irradiation, with a potential energy production of approximately 3000 kWh per year [2]. Table 1 highlights the country's solar energy production potential, with desert areas ranking among the locations with the highest average solar irradiation and temperatures worldwide. The annual sunshine duration varies from 2000 to 3900 h, while the average horizontal surface radiation falls between 3 and 5 kWh/m^2 . Furthermore, a network of 78 meteorological measurement stations, managed by the National Office of Meteorology" Office National de la Météorologie" (ONM), is mainly active through the country.

The Fig. 2 shows the annual installed capacity (in MW) of major renewable energy technologies; solar photovoltaics, hydro, wind and CSP from 2014 to 2019. Solar PV exhibits a clear and rapid growth trend, reaching approximately 400 MW in 2019,

making it the fastest growing renewable energy source during the period compared to other renewable energy sources. Solar energy is considered one of the most significant energy resources and plays a key role in addressing global energy challenges [8].

Since most solar energy is used outside buildings, architects must play a key role in the architectural integration of solar systems, particularly photovoltaic (PV) systems, ensuring that they blend harmoniously into buildings while maximizing their energy efficiency. As a mediator between technology and aesthetics, the architect must design solutions that meet functional needs while respecting the aesthetics of the building and its surrounding context. This can include integrating PV panels into facades, roofs, or even structural elements while taking into account orientation, sunlight, and shading to optimize energy production. In addition, the architect must ensure that these installations comply with construction, safety, and sustainability standards while helping to reduce environmental impact and improve occupant comfort. In this way, the architectural integration of solar PV systems makes it possible to combine energy performance and design, transforming buildings into key players in the energy transition.

The performance of solar photovoltaic (PV) systems is commonly assessed through technical, economic, and environmental criteria. Numerous studies have investigated the viability and efficiency of solar PV technologies under diverse climatic conditions globally [9–17]. Based on these insights, this study presents a projection of the potential development of solar PV in the Algerian market.

In this context, our study focuses on the solar car park in Jijel, Algeria, offering a detailed insight into the utilization of PV solar energy in structures. By examining the environmental and economic dimensions, this research aims to provide crucial information to support decision-making regarding the integration of solar energy into infrastructure. The findings of this study may also hold relevance in a broader context, enhancing the understanding of the implications of harnessing solar energy in buildings.

3 Environmental impact analysis of PV systems

Scientific and policy discourse has long emphasized the detrimental environmental consequences of conventional energy systems. In stark contrast, non-conventional energy sources particularly renewable technologies are often associated with a 'low-carbon' or 'environmentally benign' reputation, largely due to their reduced ecological footprints and lower emissions profiles [18]. Most environmental analyses and regulatory instruments have focused on conventional gaseous pollutants, including SO₂, NO₂, CO₂, particulates, and CO [19]. See Table 2.

3.1 Air pollution

The manufacturing of current-generation photovoltaic (PV) systems is energy-intensive, particularly for polycrystalline and monocrystalline modules, which require significant amounts of bulk materials. Thin-film modules, on the other hand, have a lower primary energy demand per watt compared to a-Si PV modules (which are also thin films). For polycrystalline and monocrystalline modules, manufacturing-related emissions are estimated at 2.757–3.845 kg CO₂/kWp, 5.049–5.524 kg SO₂/kWp, and 4.507–5.273 NO_x/kWp [20]. In addition, cadmium emissions from CdTe module production are minimal, representing approximately 0.001% of the cadmium used, which corresponds to about 0.01 g per GWh of energy produced. Furthermore, cadmium, a by-product of zinc production,

Table 2 Thresholds for regulated pollutants as defined by the European Union’s ambient air quality standards [19]

Pollutant	EU limit
CO	30 mg/ m ² , 1 h
NO ₂	200 µg/ m ² , 1 h
O ₃	235 µg/ m ² , 1 h
SO ₂	250–350 µg/ m ² , 24 h 80–120 µg/ m ² , annual
PM10	250 µg/m ² ,24 h 80 µg/ m ² , annual
SO2 + PM10	100–150 µg/m ² , 24 h 40–60 µg/m ² , annual
Pb	2 µg/m ² , annual
Total suspended particulate	26 µg/m ² , 24 h
HC	160 µg/m ² , 3 h

can either be utilized for beneficial purposes or inadvertently released into the environment [21]. The emissions generated by transporting PV modules are minimal compared to those produced during manufacturing. Transport-related emissions account for only 0.1–1% of the emissions associated with production.

During standard operation, photovoltaic (PV) systems do not emit gaseous or liquid pollutants, nor do they release radioactive materials. However, copper indium selenide (CIS) and cadmium telluride (CdTe) photovoltaic modules, which incorporate trace amounts of toxic materials, may pose a minor environmental risk if a fire within the array triggers their release into the surrounding environment [22].

Greenhouse-gas (GHG) emissions are most commonly expressed in kilograms of CO₂ equivalent (kg CO₂eq), a metric that represents the weighted sum of emissions, including CO₂, CH₄, and NO₂. A value of 0.98 kg CO₂eq/kWh was used by Nawaz and Tiwari [23] to represent electricity generation from coal. Other researchers, including [24], analyzed air emissions by comparing grid supply with grid-connected photovoltaic systems, as well as with off-grid (stand-alone) photovoltaic systems in Australia. The grid-connected photovoltaic system was found to be the optimal choice in terms of lifecycle emissions among the three analyzed cases. Furthermore, utilized a CO₂ equivalent value of 0.8 kg/kWh [25] for grid electricity and reported a consistent energy payback time (GPBT) of ~4 years for multi-crystalline PV systems, independent of the installation typology employed.

3.2 Land use

In this research, the photovoltaic (PV) power generation system is incorporated into the parking lot’s roof, thereby negating the necessity for extra land use.

3.3 Visual impacts

The visual impact of photovoltaic (PV) systems is is primarily influenced by the type of installation and its surrounding environment. Photovoltaic installations in environmentally sensitive areas may result in substantial visual impacts, challenging the preservation of natural scenery. Conversely, when PV modules are integrated into the facades of modern buildings, they may enhance the aesthetic appeal. However, this effect is less favorable when applied to historic buildings or structures with cultural significance, where such integration might conflict with the architectural value [26].

Effective architectural designs strive to reduce the possible effects of photovoltaic (PV) systems on visual appeal and the aesthetics of buildings, especially by incorporating them into various structures. The adoption of PV as a building envelope component in commercial architecture reveals new avenues for harmonizing renewable energy systems with architectural aesthetics, providing architects and their clients with unique design opportunities. Moreover, progress in multi-functional PV facades, which blend visual attractiveness with practical features like shading and heat management, has played a significant role in the advancement of architectural expression [27].

3.4 Depletion of natural resources

The production of photovoltaic modules, especially polycrystalline and monocrystalline types, is highly energy-intensive. However, the manufacturing process continues to require minimal quantities of scarce elements like indium (In), tellurium (Te), and gallium (Ga), in addition to restricted amounts of the toxic substance cadmium (Cd) [26].

3.5 Waste management

Autonomous energy systems, while viable, currently occupy a small fraction of the global market for power generation infrastructure [28], it is crucial to examine the health impacts of the chemical substances contained in batteries. Life cycle analyses of batteries used in stand-alone PV systems reveal that these components contribute significantly to environmental effects resulting from their relatively limited lifespan and the presence of heavy metals. Additionally, battery production requires substantial amounts of energy and raw materials. Implementing a module-recycling scheme, as suggested by Fthenakis [29], could help mitigate these issues.

4 Methodology

4.1 System description

4.1.1 Building description

Our project is located downtown Jijel, Algeria (colonial triangle) Fig. 3, within the E1 climate zone [30]. It consists of a simple structure formed by two intersecting rectangles. The parking facility is designed vertically, creating a physical and visual connection with its surroundings. Prominent within the landscape, it features a total of six levels dedicated to parking, with a capacity of 600 spaces. Additionally, the presence of a terrace offering panoramic views of Jijel is a major asset of this parking structure.

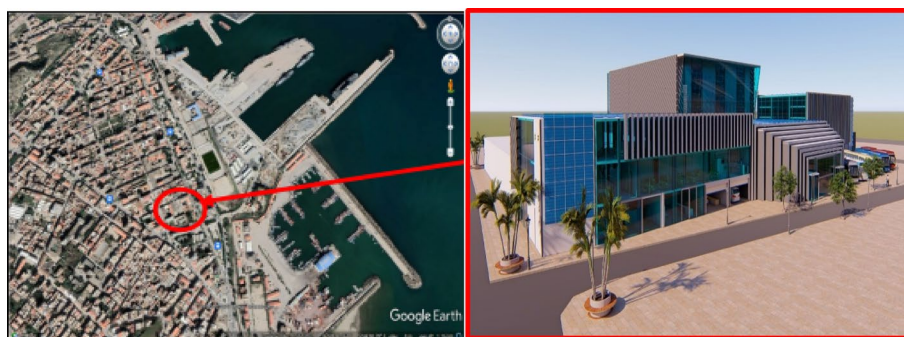


Fig. 3 Location and 3D visualization of the parking project in Jijel

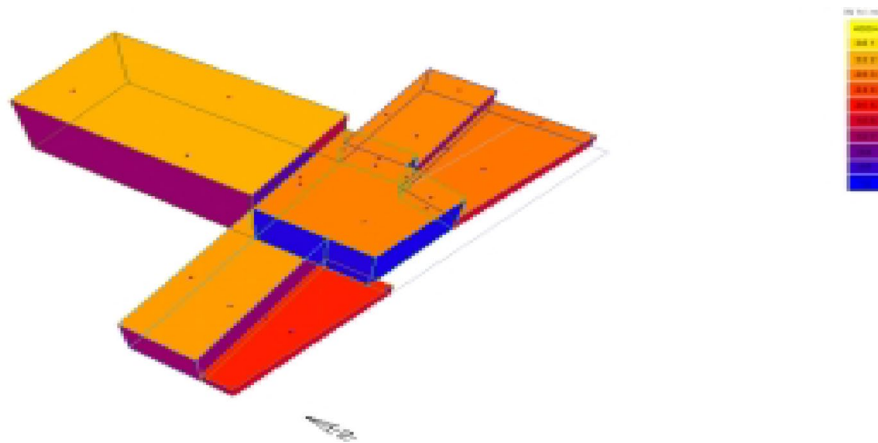


Fig. 4 Shading analysis by ECOTECT



Fig. 5 PV system integrated into rooftop building

4.1.2 Photovoltaic system description

This study explores a photovoltaic installation placed on the roof of an automated parking facility in Jijel, Algeria, with a capacity of 502 kWp (determined after a simulation using PVsyst software). This capacity enables us to cover all our electricity needs. This grid-connected system operates (no storage is considered) without any shading of the photovoltaic modules, as confirmed by a shading analysis performed using ECOTECT software Fig. 4. All modules are oriented south with a 30° tilt angle. This improves the accuracy of the sizing procedures and optimizes the performance of the PV solar energy [31], as shown in “Fig. 5”. The system comprises 1288 monocrystalline photovoltaic modules, 92 parallel strings*14 in series. Furthermore, the photovoltaic system includes three inverters, each rated at 165 kW, which are responsible to convert the direct current (DC) generated by solar panels into alternating current (AC) electricity. This AC electricity is utilized to power various electrical devices within the car park and is also fed into the power grid for the sale. Due to daily or seasonal variations in environmental conditions, such as solar irradiation and temperature, it is necessary to install three MPPT devices (9.6 A, 40.7 V). These controllers aim to optimize photovoltaic performance by tracking and adjusting voltage-current parameters in real time, ensuring sustained operation at the optimal power output level (MPPT).

Simulation results indicate that thermal losses due to elevated operating temperatures are primary efficiency concern for the grid-connected PV system. Additional losses arise

from DC wiring (1.5%), module mismatch (2%), and angle of incidence effects. String mismatch and module quality have minor impacts, with losses of 0.1%. Furthermore, the uncertainty associated with long-term degradation over a 30-year period is estimated at 0.4% [32].

The accuracy of photovoltaic system simulation heavily depends on the quality of weather data used. Therefore selecting representative climate data is an essential preliminary step in such studies. In this research, the annual energy output of BIPV system was simulated using hourly typical Meteorological year (TMY) data for Jijel, which includes solar radiance and ambient temperature values recorded at hourly intervals. This dataset was obtained from the Solcast database organization, which may be reached at (<https://www.solcast.com>) [33].

Each PV module has a rated peak power output of 390 Wp. The main technical specifications of the monocrystalline PV module, recommended for use in Mediterranean regions [34, 35], are detailed in Table 3.

To estimate the electrical energy produced by a solar panel in real-world conditions, including variations in sunlight and climate, the following equation is used [36, 37]:

$$E_{PV} = P_{STC} [1 + \beta_p (T_{cell} - T_{STC})] \frac{H_t}{H_{STC}} \quad (1)$$

where T_{STC} and T_{cell} are the photovoltaic cell temperatures under standard test and real operating conditions ($^{\circ}\text{C}$), β_p is the power temperature coefficient ($\%/^{\circ}\text{C}$), and H_{STC} and H_t are the solar irradiance levels under standard and actual conditions, respectively. A key challenge lies in identifying a suitable empirical equation to accurately estimate T_{cell} . For example [38]:

$$T_{cell} = T_{\infty} + 7.8 \times 10^{-2} H_t \quad (2)$$

4.2 Energy payback time (EPBT) estimation

A key parameter in assessing the environmental footprint of solar PV technologies is the energy payback time (EPBT). This refers to the duration a system must operate to offset the amount of primary energy consumed during its production, operation, and maintenance [39]. The energy payback time (EPBT) is calculated by Eq. (3) [40]:

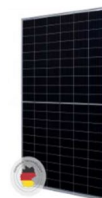
$$EPBT = \frac{E_{input}}{E_{output}} \quad (3)$$

where E_{input} is the Total Energy Input (kWh); E_{output} is the annual energy output of the system, kWh.

Table 3 Main characteristics of PV module

Rated power	390
Solar cell type	Mono-crystalline silicon
Number of solar cells	120
Module dimension	1754 mm × 1038 mm × 30 mm
Module efficiency ^a	20.32%
Frame material	Aluminum
Module weight	18.6 kg

a: figure at standard test conditions (Irradiance 1000 W/m², Cell temperature 25 $^{\circ}\text{C}$ and air mass of AM1.5).



4.2.1 Total energy input estimation

Input energy denotes the total energy utilized during the entire life cycle of a system, including the extraction of primary resources E_P , manufacturing processes E_M (The photovoltaic (PV) module is not the only element to consider, although it contributes to the majority of energy production. Critical ancillary components including inverters, electrical wiring systems, and electronic control units should not be overlooked in comprehensive system assessments, as well as mechanical components, which include mounting materials and structures [41]), transportation E_T , installation, operation and maintenance $E_{I,M}$, also decommissioning and disposal E_D . The total energy input is defined by Eq. (4):

$$E_{input} = E_P + E_M + E_T + E_{I,M} + E_D \quad (4)$$

4.2.2 Annual energy output estimation

Local environmental factors including hourly solar irradiance and ambient temperature variations significantly determine the annual energy output of PV system, the efficiency of the PV modules, and the design of the system (including its orientation and whether it is grid-connected or off-grid) as shown in Table 4.

4.3 Greenhouse-gas payback time (GPBT) estimation

In basic terms, it is the duration needed for the system to “offset” its initial carbon footprint by producing clean energy and decreasing emissions that would have otherwise resulted from fossil fuel use. The greenhouse-gas (GPBT) is given by Eq. (5) [40]:

$$GPBT = \frac{GHG_{input}}{GHG_{output}} \quad (5)$$

where GHG_{input} : The greenhouse-gas emissions that were released during its production, transportation, and installation. GHG_{output} : Yearly carbon dioxide equivalent

Table 4 Daily energy consumption

Zone	Machine	Number	Operating time (hour/day)	Power (W)	Consumption/day (Wh)
Administration	Lamps	30	10	18	5400
	Computer	4	8	80	2560
	Split-system	3	8	1300	436,800
	Printers	3	8	10	240
	Landline telephones	3	8	8	192
	Refrigerator	1	24	150	3600
	(Portable charger)	6	2	6	72
Station zone	Car elevator	8	10	14,000	1,120,000
	Car wash	1	16	45,000	720,000
	Barriers	1	14	80	1120
	LED Floodlights	15	14	30	6300
	Elevator	2	8	4000	64,000
2Cafe + restaurant	Lamps	35	8	18	5040
	Refrigerator	5	24	150	25,200
	Cooking machine	4	10	200	8000
	TV	2	10	100	4000
	Split-system	1	8	1800	14,400
	Split-system	1	8	2400	19,200
	(Portable charger)	7	2	6	84
Total					2,565,808 (Wh)

emissions (kg CO₂eq) from the local power plant corresponding to the energy produced by the PV system.

5 Findings and discussions

5.1 Availability analysis of surfaces

A significant part of this roof surface, 2550 m², is dedicated to the electrical needs of the parking lot. This ensures the energy autonomy required to power the lighting, automated systems and other equipment. A further 2000 m² is available for the production of surplus electricity Fig. 6. This additional energy, fed into the Sonelgaz power grid, offsets energy consumption during night-time periods. As the system is connected to the grid, no energy storage is required, simplifying installation while generating additional income for the operator.

The allocation of 53% of the roof's surface area for energy self-sufficiency demonstrates the effective sizing of the photovoltaic system to meet the parking lot's energy requirements completely. This approach enhances the sustainability of the project while making efficient use of available resources.

5.2 Energy payback time of the PV system (EPBT)

5.2.1 Energy input of the PV system

Producing 1 kg of MG-Si requires 20 kWh of energy [42–44]. For the production of 1 kg of EG-Si, 100 kWh of energy is consumed, with a silicon loss rate of 90% during the process. The Czochralski process for monocrystalline silicon production consumes 290 kWh/kg of energy and results in a 72% material loss. Considering that 1.448 kg of silicon is required to fabricate 1 m² of photovoltaic cells, the embodied energy from purification and processing stages for this area is computed using Eq. (6):

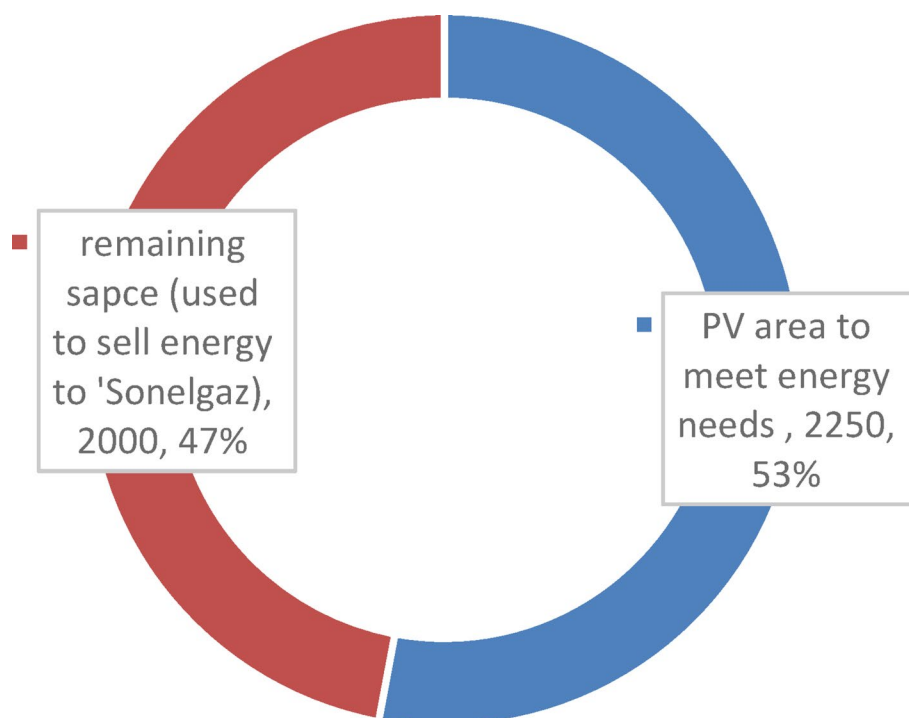


Fig. 6 Rooftop area allocation for PV system and remaining space

$$E_p = (290 \times 1.448) + \left(\frac{1.448}{0.72} \right) \times \left(100 + \left(\frac{20}{0.9} \right) \right) = 665.72 \left[\frac{kWh}{m^2} \right] \quad (6)$$

The energy required for the manufacturing processes is 120 kWh/m² to be sliced into wafers [21], in addition to 190 kWh/m² to obtain a finished photovoltaic module [45]. For photovoltaic systems integrated into rooftops, the energy needed to produce the supporting structure [46] amounts to 200 kWh/m². Additionally, 33 kWh/m² is consumed as embodied energy during the production of inverters [21]. Regarding the energy consumed for transportation, due to the lack of information or its absence for Algeria, we estimated it to be 2%, based on a study conducted by the Lu researcher [40]. The operational energy demand, encompassing maintenance activities, electronic systems, cabling, and ancillary components [40], amounts to 125 kWh/m².

Consequently, the total embodied energy of the 502 kW grid-connected photovoltaic system is 3,184,496 kWh. The pie chart illustrates that a significant portion of the embodied energy in photovoltaic systems is attributed to the extraction and processing of raw materials, accounting for 49%, as well as manufacturing components like modules (14%) and support structures (15%). The operational phase and maintenance has a minor energy demand (9%) Fig. 7.

5.2.2 Energy output of the PV system

For the 502 kW photovoltaic system installed on the roof, oriented southward at a 30° angle of inclination, it is designed to meet the parking lot's projected electrical demand of 2565.808 kWh/day (see Tab. (4)), the annual energy production (AC electricity) is 936,519.92 kWh. The system's Energy Payback Time (EPBT) is estimated at 3.4 years. This value falls within the typical range reported in the literature for photovoltaic (PV) systems, which generally have an energy payback time (EPBT) of less than 7.1 years [9, 43, 47, 48], significantly shorter than the typical lifespan of photovoltaic (PV) modules, which generally ranges from 20 to 30 years.

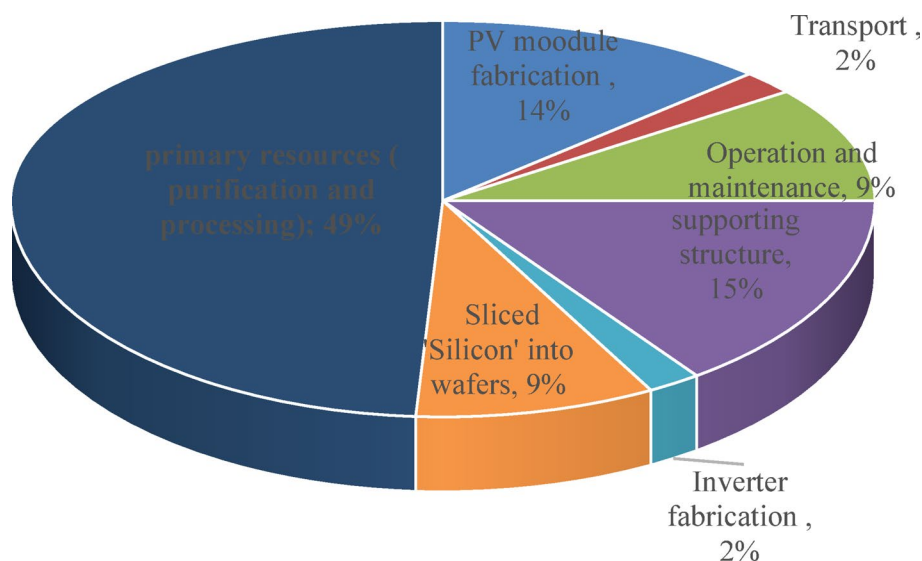


Fig. 7 Proportional distribution of embodied energy in a rooftop building-integrated photovoltaic (BIPV) system

5.3 Greenhouse-gaz payback time of PV system (GPBT)

In Jijel, there is only one public utility company called Sonelgaz, responsible for supplying electricity to the building sector. Thanks to the availability of natural resources, particularly natural gas (which leads to dependence on fossil fuels), Sonelgaz operates the Achouat power plant to supply electricity to the city of Jijel. The PV system's annual AC power output is estimated at 936,519.92 kWh. The corresponding reduction in CO₂ emissions from the power station is: $936,519.92 \times 0.8 \text{ KgCO}_{2\text{eq}}/\text{kWh} = 749,215.93 \text{ KgCO}_{2\text{eq}} = 749.216 \text{ T eq CO}_2$. For the greenhouse gases embodied during the fabrication of the cells, a value [25] of 463 kg CO_{2eq}/m² was utilized for the production of photovoltaic modules. In terms of the balance of system (BOS) components that influence the calculation of embodied energy for rooftop installations, an estimate of 6.1 kg CO_{2eq}/m² was assigned to array supports and cabling [49], while inverters were attributed [23] 125 kg CO_{2eq}/m².

Greenhouse gas emissions associated with transportation and disposal are not included in this analysis. The pie chart indicates Fig. 8. that the majority of greenhouse gas emissions are concentrated in the fabrication processes (78%), which should be the primary focus for emission reduction strategies. Addressing emissions from installation processes can also contribute to overall improvements. The negligible impact of inverters suggests limited opportunities for emission reductions in this category.

The carbon footprint attributable to the manufacturing process and installation of the photovoltaic system are estimated to be 1392.665 TeqCO₂. Consequently, The PV system's greenhouse gas payback time (GPBT) is calculated to be approximately 1.9 years. It is typically reported as a value in the literature for BIPV systems, which generally have GPBT payback time between 0.4 and 1.9 years [50], and up to five years [9, 43].

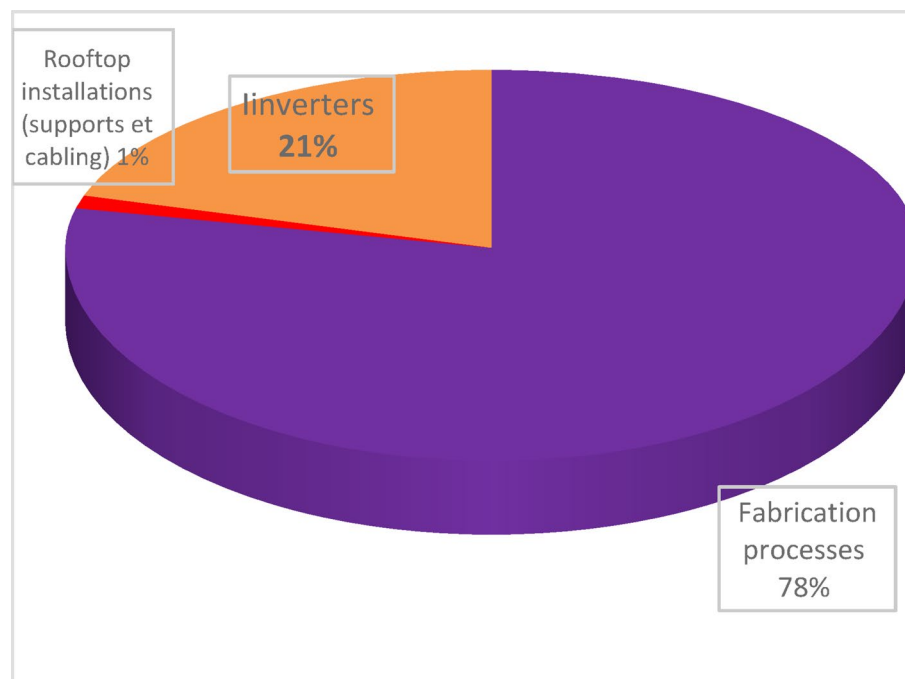


Fig. 8 Percentage contribution of greenhouse gas emissions from PV system

Table 5 Estimation cost of the photovoltaic (PV) system to meet the energy needs of a parking facility

	Price (\$/unit)	Number	Total price (\$)
Pv panel	163	1288	209,944
inverters	4137,54	3	12,412.62
Mppt	3149,48	3	9448.44
Investment cost			231,805.06

5.4 Economic analysis

From an economic standpoint, calculating some economic indicators is considered very important to demonstrate the effectiveness of projects to investors. Among these important indicators are: The cost of generated energy unit (LCOE) and the payback period. The cost of environmental damage produced by carbon dioxide (CCO₂) may be used to compute LCOE using the following Eq. [36]:

$$LCOE = \frac{\left(\frac{r(1+r)^n}{(1+r)^n - 1} \right) \times C + C_{O\&M} - C_{CO_2}}{E_t} \quad (7)$$

where C the capital cost of the system in \$, $C_{O\&M}$ denotes the cost of operation and maintenance (\$/year), E_t is annual energy produced by the system (kWh/year), n is the device lifetime (30 years), r the annual inflation rate.

The cost of environmental damage (CCO₂) caused by CO₂ gas can be calculated by the following Eq. 4 [5].

$$C_{CO_2} = EF_{CO_2} \times E_t \times \emptyset_{CO_2} \quad (8)$$

where: EF_{CO_2} represents the CO₂ emission factor of the electric power generation system (kg CO₂/kWh), $\emptyset_{CO_2}$ represents the carbon social cost (\$/ton CO₂), which may be considered as \$ 70/ton CO₂ [51].

5.4.1 Payback period

The payback period is a crucial economic indicator used to evaluate the financial viability of photovoltaic systems. It represents the number of years necessary to realize the total investment [52].

Table 5 presents the estimated investment cost for the solar system. However, the following components have not been included in this estimation: installation of the photovoltaic (PV) system, design and engineering costs, electrical wiring expenses, Operating costs, including maintenance, insurance, and metering fees.

The table outlines the main investment costs associated with the photovoltaic (PV) system designed to meet the energy demand of the parking lot. The largest share of the total cost is attributed to the PV panels, which amount to 209,944 \$ for 1288 units. This highlights the dominant role of solar modules in the overall system expenditure, compared to other components such as inverters and MPPT controllers.

Overall, the total initial investment for the PV system reaches approximately 231,805.06 \$, excluding additional expenses such as installation, design, wiring, and operation fees. A payback period can be calculated by using Eq. (9) [36, 53].

$$\text{payback period (years)} = \frac{\text{Investment Costs}}{\text{Annual dollar Savings}} = \frac{231805.06\$}{28114.26\$} = 8.24\text{years} \quad (9)$$

The result indicates that the photovoltaic (PV) system will recover its initial investment cost in approximately 8.24 years, based solely on the annual energy savings generated. This value falls within the typical range reported in the literature for grid-connected solar PV systems, which generally have payback periods more than 5 years [9, 54–58], depending on location, system size, electricity tariffs, and available incentives.

It is important to note that this calculation was based on the subsidies tariffication of energy in our country with the value of (0.03 \$/kWh), nevertheless, the obtained payback period suggests that the PV system installed to meet the parking lot's energy needs is economically viable, especially if the system lifetime exceeds 25 years, as is common for modern photovoltaic installations.

5.4.2 Energy Sales to Sonelgaz estimation

To transform the building into an energy producer, we are utilizing the remaining 2000 m² of roof space to install a photovoltaic (PV) system. This installation will generate electricity for sale to Sonelgaz (the public electricity distribution company). The available surface area can accommodate approximately 1010 solar panels, providing a total capacity of 390 kW and an estimated annual output of 734,525 kWh.

Since the building relies on Sonelgaz's electricity for its night-time operations, it is estimated that about one-third of the energy produced approximately 244,842 kWh will be used for its own night-time consumption. The remaining two-thirds, equating to around 489,683 kWh per year, will be allocated for sale.

With a tariff of 0.03 \$/kWh set by Sonelgaz, the annual revenue from energy sales is projected to reach approximately 14,658,50 \$. Additionally, leveraging photovoltaic solar panels as an energy source will result in further savings of approximately 28,114.26 \$ on the building's energy expenses, enhancing both the economic and environmental advantages of the project.

6 Conclusions

The integration of photovoltaic (BIPV) technology into building design exemplifies a transformative approach to sustainable urban development, delivering significant environmental and economic benefits. This project demonstrates a lifespan energy savings of 936,519.92 kWh/year, directly reducing reliance on fossil fuel-dependent energy systems. With an energy payback time (EPBT) of 3.4 years and a greenhouse gas payback time (GPBT) of 1.9 years, the system achieves rapid financial and ecological returns, underscoring its viability as a cost-effective solution for energy transition. Despite the subsidized tariffication of energy in Algeria by a value of 0.03 \$/kWh, the payback period resulted shows an acceptable value of 8.2 years.

Beyond the technical performance, the design of the rooftop PV array and terrace integration highlights the critical role of architects in harmonizing sustainability with architectural aesthetics and social value.

Architects play a dual role as visionary designers and sustainability advocates, bridging the gap between technological innovation and functional urban spaces. In this project, their expertise ensured that PV systems were not merely retrofitted but integrated holistically into the building's rooftop. By optimizing solar panel orientation, material selection, and spatial planning. Architects maximized energy generation while preserving the building's visual appeal and urban context.

This case study serves as a blueprint for future urban infrastructure projects, illustrating how architectural innovation can drive renewable energy adoption in the built environment. In Algeria's tertiary sector where commercial and institutional buildings account for a significant energy footprint such projects are vital for achieving national sustainability targets. By prioritizing solar PV integration, architects can reduce embodied and operational carbon emissions while fostering energy resilience.

In conclusion, this initiative not only validates the technical and economic viability of PV systems but also reaffirms that architectural design is a cornerstone of energy sustainability. By championing solar integration, architects can transform buildings into climate-positive assets, paving the way for a decarbonized future where functionality, aesthetics, and environmental responsibility converge.

Author contributions

Ouahid Halloufi, Abdenacer Kaabi, Maamar Hamdani, and Takele Ferede Agajie: Conceptualization, Methodology, Software, Visualization, Investigation, Writing- Original draft preparation, Project administration, Funding Acquisition. Antar Leflef, Mohamed Aymen Gamraoui, Mohamed Kamal Cherier, and Abdelhamid Boualit: Data curation, Validation, Supervision, Resources, Writing—Review and Editing.

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Availability of data and materials

The data used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

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